



OBJECT ORIENTED PROGRAMMING (OOP)

MR. PREM KUMAR

LECTURER

DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

AGENDA

- STATIC KEYWORD
- STATIC VARIABLE
- STATIC METHOD
- STATIC BLOCK
- INNER CLASSES

STATIC KEYWORD

- The **static keyword** in **Java** is used for memory management mainly.
- We can apply static keyword with **variables**, methods, blocks and **nested classes**.
- The static keyword belongs to the class than an instance of the class.

STATIC KEYWORD CAN BE:

- The static can be:
 1. Variable (also known as a class variable)
 2. Method (also known as a class method)
 3. Block
 4. Nested class

I) JAVA STATIC VARIABLE

- The static variable can be used to refer to the common property of all objects (which is not unique for each object), for example, the company name of employees, college name of students, etc.
- The static variable gets memory only once in the class area at the time of class loading.
- Example of Java static variable using counter variable

JAVA STATIC METHOD

- If you apply static keyword with any method, it is known as static method.
- A static method belongs to the class rather than the object of a class.
- A static method can be invoked without the need for creating an instance of a class.
- A static method can access static data member and can change the value of it.

EXAMPLE OF STATIC METHOD

```
class Student{
    int rollno;
    String name;
    static String college = "ITS";
    static void change(){
        college = "BBDIT";
    }
    Student(int r, String n){
        rollno = r;
        name = n;
    }
    void display(){System.out.println(rollno+" "+name+" "+college);}
}
```

```
public class TestStaticMethod{
    public static void main(String args[]){
        Student.change();
        Student s1 = new Student(111,"Karan");
        Student s2 = new Student(222,"Aryan");
        Student s3 = new Student(333,"Sonoo");
        //calling display method
        s1.display();
        s2.display();
        s3.display();
    }
}
```

TWO MAIN RESTRICTIONS OF STATIC METHOD

1. The static method can not use non static data member or call non-static method directly.
2. this and super cannot be used in static context.

Q) WHY IS THE JAVA MAIN METHOD STATIC?

- Ans) It is because the object is not required to call a static method. If it were a non-static method, JVM creates an object first then call main() method that will lead the problem of extra memory allocation.

Q) IS THERE ANY VARIABLE, METHOD, FUNCTION WHICH CAN EXECUTED BEFORE MAIN METHOD?

- Answer is YES! It is static block
- Is used to initialize the static data member.
- It is executed before the main method at the time of classloading.

```
class A2{  
    static{System.out.println("static block is invoke  
d");}  
    public static void main(String args[]){  
        System.out.println("Hello main");  
    }  
}
```

INNER CLASSES

- A class inside another class is known as inner classes
- There are two types of inner classes
 - Member class
 - Static member class
- An inner class can:
 - Access the private members of the containing class
 - Have a private access modifiers



THANK YOU